

Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c\}$. Give an example of a function $f : A \rightarrow B$ that is neither injective nor surjective.

$f : A \rightarrow B$ would be noninjective or surjective like this:

$f : A \rightarrow B = \{\{1, a\}, \{2, a\}, \{3, a\}, \{4, a\}\}$

In this example, $a, a' \in A$ $a \neq a'$, but $f(a) = f(a')$ and every $b \in B$ does not have an $a \in A$ for $f(a) = b$

Suppose $A = \{a, b, c, d\}$, $B = \{2, 3, 4, 5, 6\}$ and $f = \{(a, 2), (b, 3), (c, 4), (d, 5)\}$. State the domain and range of f . Find $f(b)$ and $f(d)$.”

$f : A \rightarrow B$

Domain : A

Range : $\mathbb{N} f(b) = 3$

$f(d) = 5$